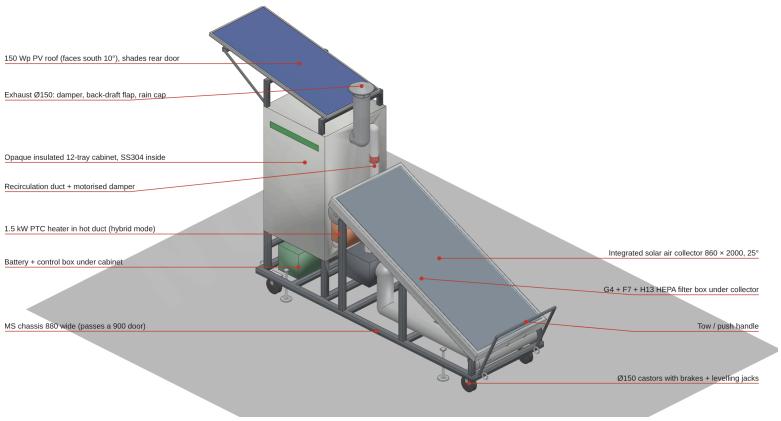
নবশক্তি প্রকৌশল

Nobo Shakti Prokushal

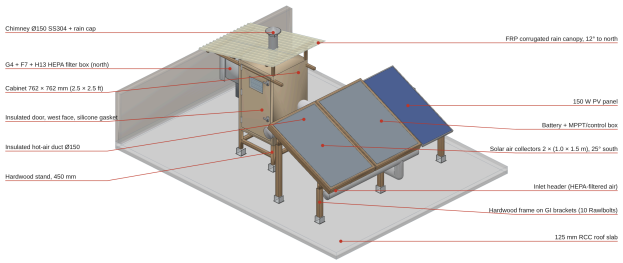
NOBOSHAKTI PROKUSHAL (NSP) · PRODUCT CATALOG 2026

Clean-Air Solar Food Dryers

HEPA-filtered, 100% solar-DC cabinet dryers for fruit, vegetables, herbs and spices — designed and built in Bangladesh.



NSP-M2 · MOBILE SOLAR-HYBRID



NSP-SD12 (M1) · ROOFTOP SOLAR

www.noboshaktiprokushal.com

+880 1714 073604 · info@noboshaktiprokushal.com

Engineering · Fabrication · Clean energy

Dhaka, Bangladesh

ABOUT NOBOSHAKTI PROKUSHAL

Engineering you can rely on, now drying food the clean way.

NoboShakti Prokushal (NSP) — নবশক্তি প্রকৌশল — began in 2007 as an electromechanical design consultancy in Dhaka. Today our team of engineers designs, fabricates and maintains mission-critical systems for more than 40 hospitals, and has delivered solar home systems, biogas plants and custom fabrication projects across Bangladesh.

2007	20+	15+	40+	200
Founded in Dhaka	Years of engineering	Engineers on the team	Hospitals served	Solar home systems, Kurigram

Why we build food dryers

Bangladesh grows mango, jackfruit, pineapple, banana, guava, tomato, chilli and moringa in abundance — and loses a large share of every harvest to spoilage. Sun-drying on open mats exposes food to dust, insects, birds and road pollution, and produces a product that cannot meet export or supermarket standards.

NSP dryers bring hospital-grade clean-air engineering to food drying: filtered, sun-heated air in a closed stainless-steel cabinet, powered entirely by the sun.

Our larger goal

These dryers are the first step toward a formal, food-grade dry-fruit and vegetable processing industry in Bangladesh — sourcing hormone-free, low-chemical and, progressively, organically grown produce through a supply network that includes farmers and organic-fertiliser producers as key partners.

What NSP brings to every dryer

- **In-house engineering** — 3D model, full drawing set and calculations for every model.
- **Clean-air know-how** from hospital HVAC and medical gas work: staged G4 → F7 → H13 HEPA filtration, positive-pressure air path.
- **Own workshop** in Khilgaon, Dhaka for steel, stainless-steel and timber fabrication.
- **Solar experience** — PV sizing, MPPT charge control and LiFePO4 storage.
- **Local service** — installation, commissioning, operator training and spare filters from Dhaka.

Led by Engr. Mostafa Shawkat Imran

Founder · Mechanical Engineer (RUET) · MBA (AUST) · Fellow, Institution of Engineers Bangladesh (F-11718)





NoboShakti Prokushal

WHY CHOOSE AN NSP DRYER

Ten reasons buyers choose NSP

Every NSP dryer is designed around one idea: **clean, gentle heat with no running fuel cost**, so the product you sell looks, tastes and keeps better.

01

HEPA-filtered clean air

Three-stage G4 → F7 → H13 HEPA filtration and a sealed, positive-pressure air path keep dust, pollen, smoke and insects out of the product — even next to a busy road.

02

SS304 food-contact surfaces

Stainless-steel mesh trays, cabinet liner and air distributor. Smooth, washable, corrosion-free and suitable for food-grade hygiene routines.

03

100% solar DC power

150 Wp PV panel, MPPT controller and LiFePO4 battery run the fan and controls. No grid, no fuel, near-zero operating cost in solar mode.

04

Indirect drying — better colour & nutrients

Food sits in an opaque cabinet, out of direct sunlight. Heated air does the work, so colour, aroma and light-sensitive vitamins are better preserved than in open sun-drying.

05

Controlled temperature

Drying air held around 50–60 °C with an adjustable exhaust damper and fan speed, avoiding case-hardening and scorching.

06

Hybrid heat for 24-hour drying M2

A 1.5 kW PTC heater with airflow interlock and over-temperature cut-out, plus air recirculation, keeps drying going at night and in cloudy or monsoon weather.

07

Mobile — dry where it grows M2

Braked castors, levelling jacks, tow handle and tie-down eyes. 880 mm wide, so it passes a standard 900 mm door and goes from orchard to shed.

08

Data-logging ready

Temperature and humidity sensing with an optional ESP32 logger gives a batch record for traceability and HACCP-style quality systems.

09

Compact, modular, one cabinet

Both models share the same 12-tray, 2.5 × 2.5 ft cabinet and spare parts, and the same cabinet is the base for NSP's future heat-pump model.

10

Made in Bangladesh, serviced locally

Designed and fabricated in Dhaka with locally available materials. Fast spares, local-language training and on-site support.

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TECHNOLOGY

Clean air in, moisture out

NSP dryers are **indirect, fan-forced solar cabinet dryers**. Outside air is filtered first, heated in a solar air collector, then pushed up through 12 stainless-steel trays and out of the chimney, carrying the moisture with it.

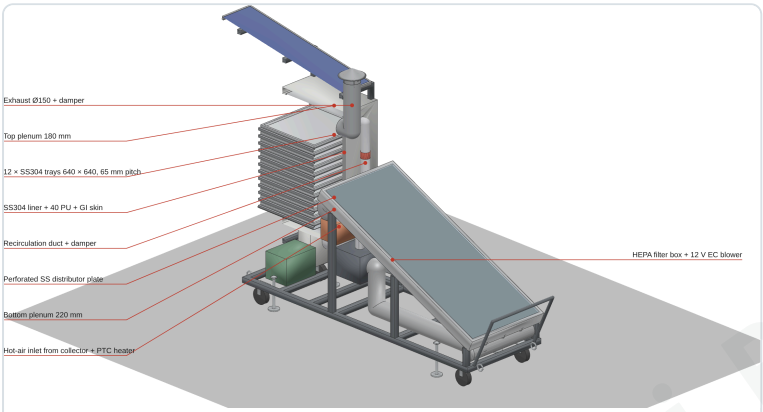
1 FILTER — Air enters a sealed filter box: G4 pre-filter, F7 fine filter, H13 HEPA.

2 BLOW — A 12 V DC EC blower pressurises the clean air path, so unfiltered air cannot leak in.

3 HEAT — The back-pass solar collector heats the air to about 50–60 °C. In M2, a PTC heater can take over.

4 DRY — A perforated SS distributor spreads air evenly up through 12 SS304 trays.

5 EXHAUST — Moist air leaves via a damper, back-draft flap and rain-capped chimney.



NSP-M2 cut-away — plenums, trays, insulated wall and hot-air path

Operating modes

MODE	HOW IT RUNS
Solar (M1, M2)	Collector heat + solar-DC fan. Daytime drying, zero fuel cost.
Solar + boost (M2)	PTC heater tops up on cloudy days; thermostat holds the set point.
Hybrid 24 h (M2)	Heater + partial recirculation overnight; damper vents humid air.
Night hold (M1, M2)	Dampers closed, battery keeps the fan on low to stop moisture re-absorption.

Hygiene by design

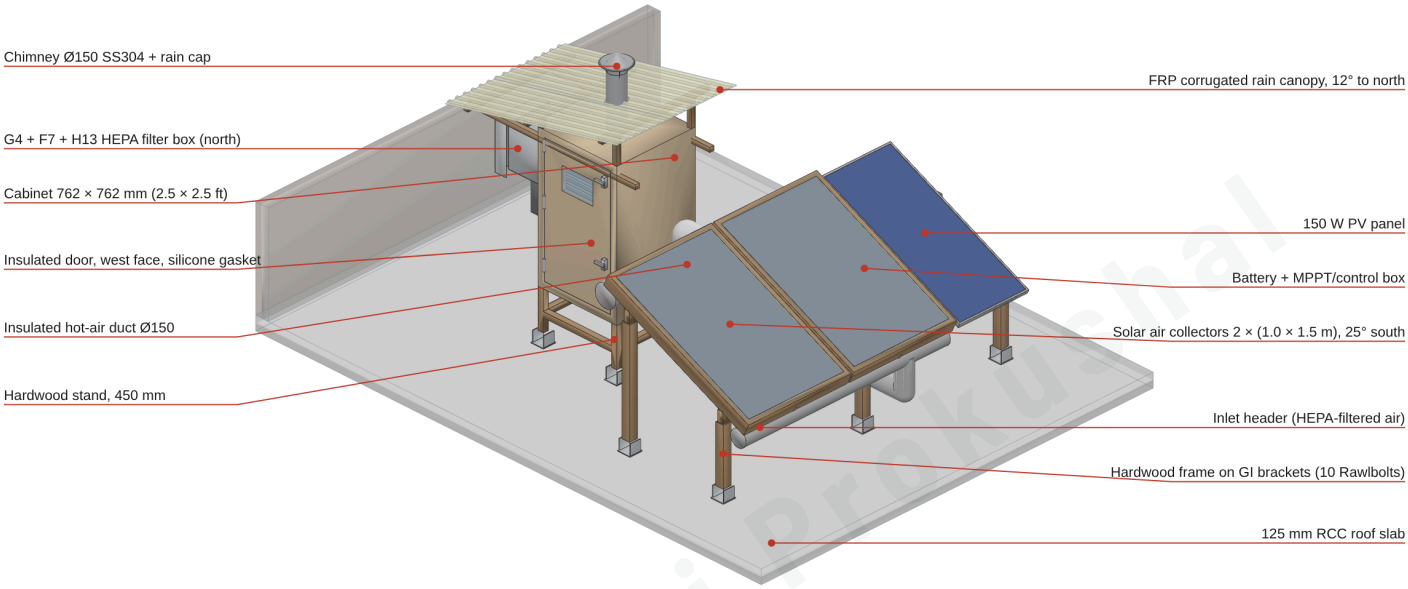
- ✓ No open-air contact between food and outside air.
- ✓ Filters changed from outside the food zone.
- ✓ Removable trays and a smooth SS liner for wet cleaning.
- ✓ Opaque cabinet: no direct UV on the product.
- ✓ Silicone door gasket; insect-proof exhaust flap.

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MODEL M1 · ROOFTOP · FIXED · 100% SOLAR DC

NSP-SD12 Rooftop Solar Cabinet Dryer

A fixed, 12-tray clean-air solar dryer for rooftops and yards. Two large solar air collectors give it the highest solar-only output in the NSP range — ideal for home producers, cooperatives and pilot processing units.



12
SS304 trays · 4.9 m^2 tray area

3.0 m^2
Solar air collector area

≈ 8 kg
Fresh slices per 2 clear days*

150 Wp
PV · runs fan & controls

Best for

- Rooftop or courtyard installation in towns and peri-urban areas
- Small brands and cooperatives starting dried-fruit production
- Dry-season drying of mango, pineapple, banana, tomato, chilli and leafy herbs

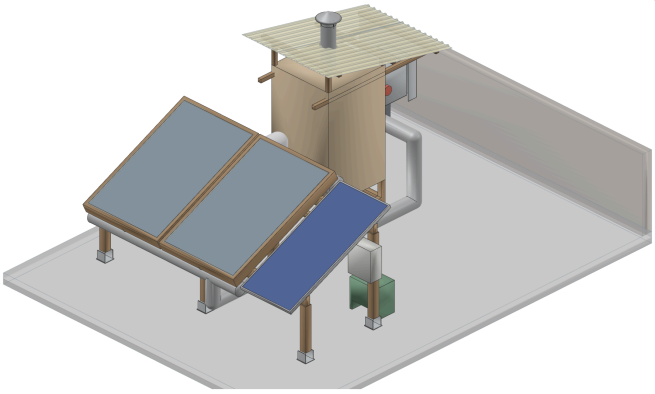
Highlights

- Anchored with 10 Rawlbolts to an RCC roof; designed for monsoon wind
- Corrugated FRP rain canopy over a sealed cabinet top
- Timber + BWR plywood shell, rockwool insulation, SS304 liner: easy to build and repair locally

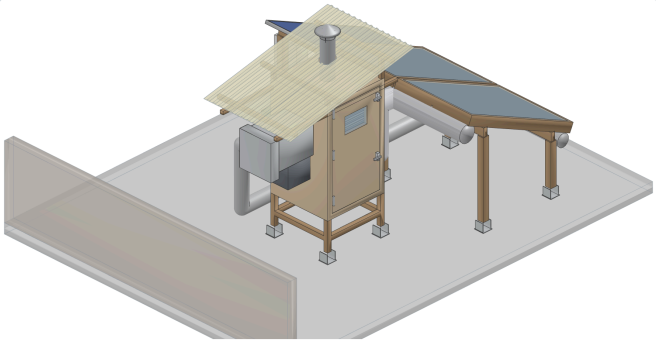
* Design estimates for dry-season conditions in Dhaka; actual output depends on crop, slice thickness and weather. Figures will be confirmed in NSP commissioning trials.

M1 · 3D VIEWS

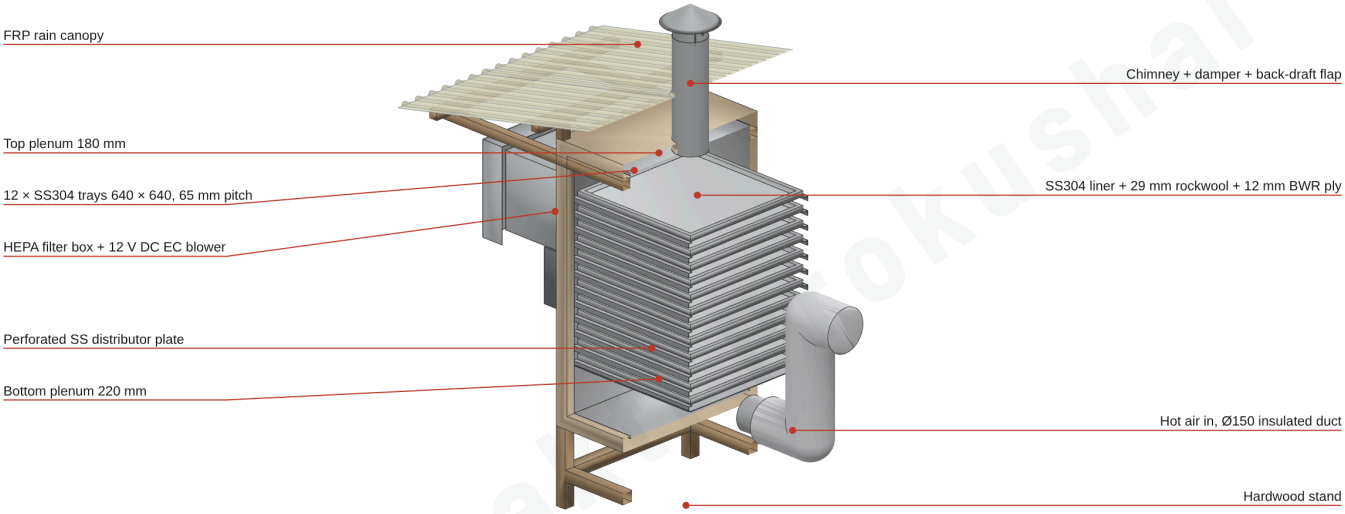
Every angle, every part



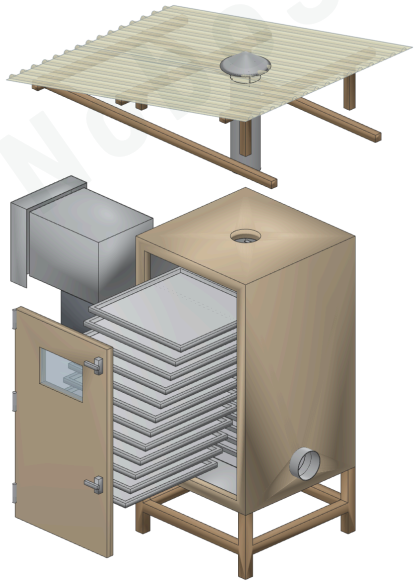
South-east view — collectors and PV face south



North-west view — filter box, door and canopy



Cut-away — air path from HEPA filter box through collectors to trays and chimney



Exploded view — trays, liner, shell, plenums



Plan view — compact 4.0 × 4.5 m rooftop zone

M1 · TECHNICAL SPECIFICATIONS

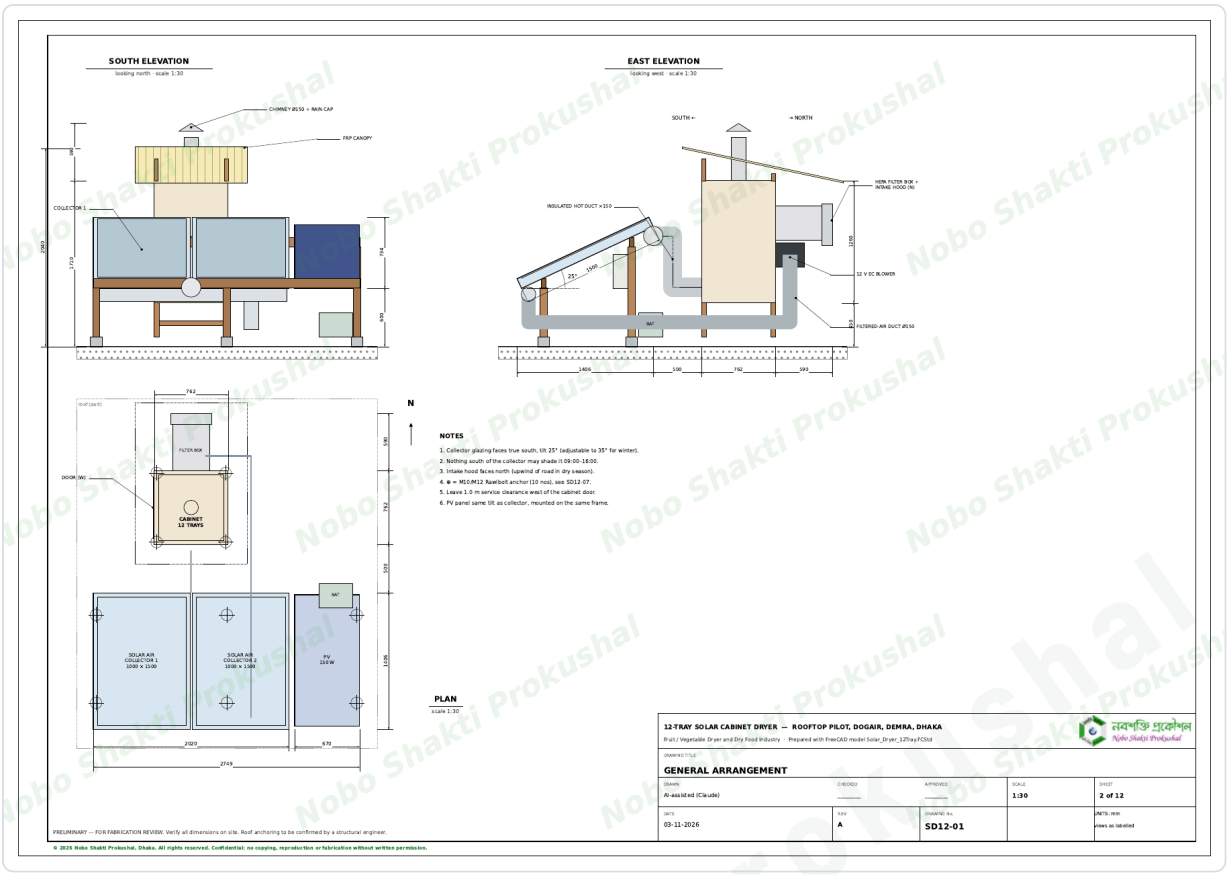
NSP-SD12 technical data

ITEM	SPECIFICATION
GENERAL	
Type	Indirect, fan-forced, once-through solar cabinet dryer with filtered positive-pressure air path
Installation	Fixed — rooftop (RCC slab) or ground; collectors face true south
Footprint zone	≈ 4.0 × 4.5 m including service access
DRYING CABINET	
Cabinet plan size	762 × 762 mm (2.5 × 2.5 ft); body height 1,260 mm on 450 mm hardwood stand
Wall construction	42 mm: 12 mm BWR plywood + 29 mm rockwool + SS304 liner
Trays	12 × SS304 mesh trays, 640 × 640 mm, 65 mm pitch — 4.9 m^2 total
Door	Insulated, west face, silicone gasket, two compression latches
Air distribution	Bottom plenum with perforated SS distributor plate; top plenum to chimney
Rain protection	Separate semi-transparent corrugated FRP canopy, 12° fall to north
SOLAR THERMAL	
Collector	2 × 1.0 × 1.5 m back-pass solar air collectors (3.0 m^2), parallel flow
Tilt / orientation	25° (adjustable 25–35°), facing south
Drying air temperature	≈ 50–60 °C at tray inlet on clear days (design)
AIR HANDLING & FILTRATION	
Filtration	G4 pre-filter → F7 fine filter → H13 HEPA (305 × 305 mm) in a sealed 380 × 480 × 350 mm filter box with clean plenum
Fan	12 V DC EC blower, ≥ 150 m^3/h free air, ≥ 400 Pa
Operating airflow	60–120 m^3/h (speed controlled)
Ducts / exhaust	Insulated Ø150 hot-air duct; Ø150 SS304 chimney with damper and rain cap
ELECTRICAL (100% SOLAR DC)	
PV module	150 Wp, on its own frame next to the collectors
Charge controller	20 A MPPT, 12 V
Battery	12 V 40 Ah LiFePO4 (fan night-hold and controls)
Controls & sensing	Fan speed control, DS18B20 temperature sensors; optional ESP32 data logger
STRUCTURE & PERFORMANCE	
Frame	Seasoned hardwood on GI brackets; anchored with 10 × M10/M12 Rawlbolts
Design batch	≈ 8 kg fresh sliced fruit per 2 clear dry-season days
Running cost (solar mode)	No fuel, no grid power; filters and maintenance only

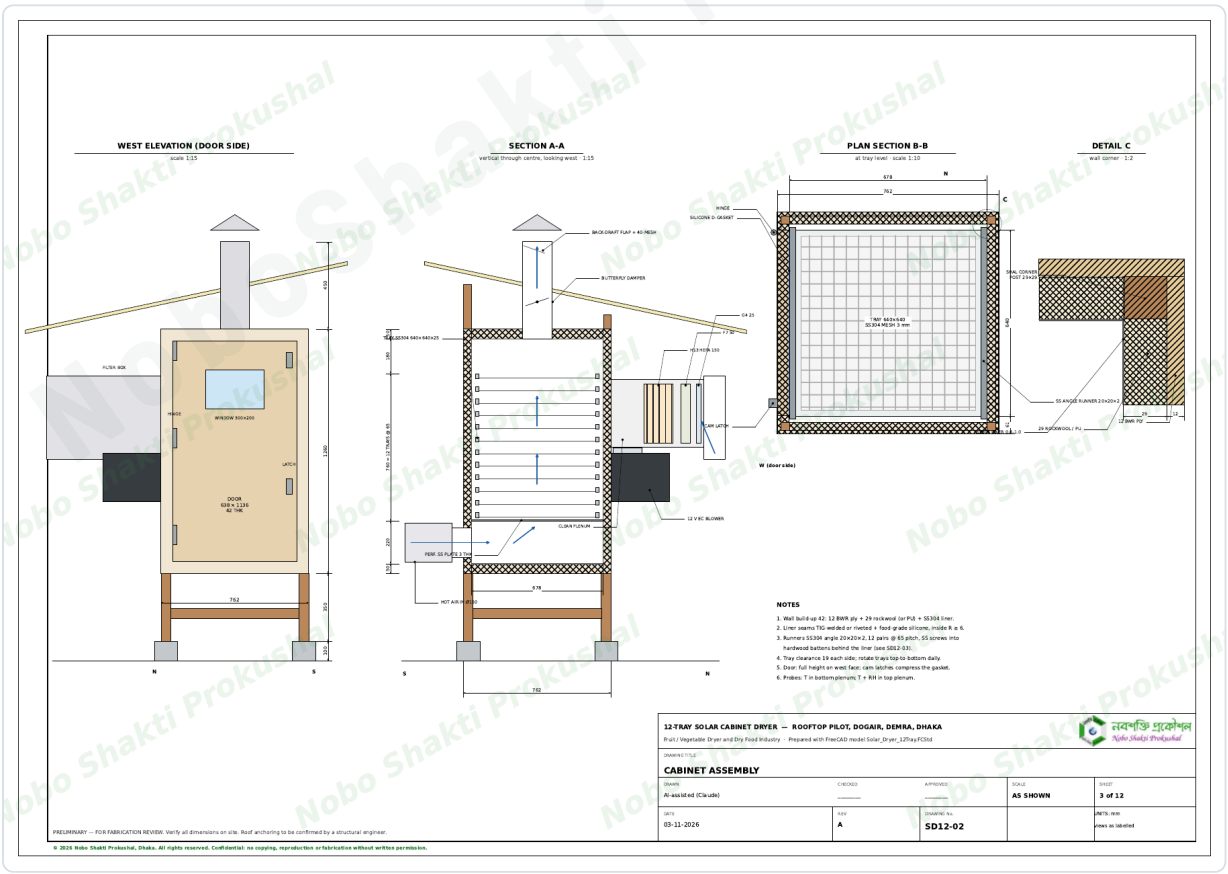
Specifications are for the rev A design and may change as the product is improved. Structural anchoring must be confirmed for each roof by a qualified engineer.

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M1 · ENGINEERING DRAWINGS

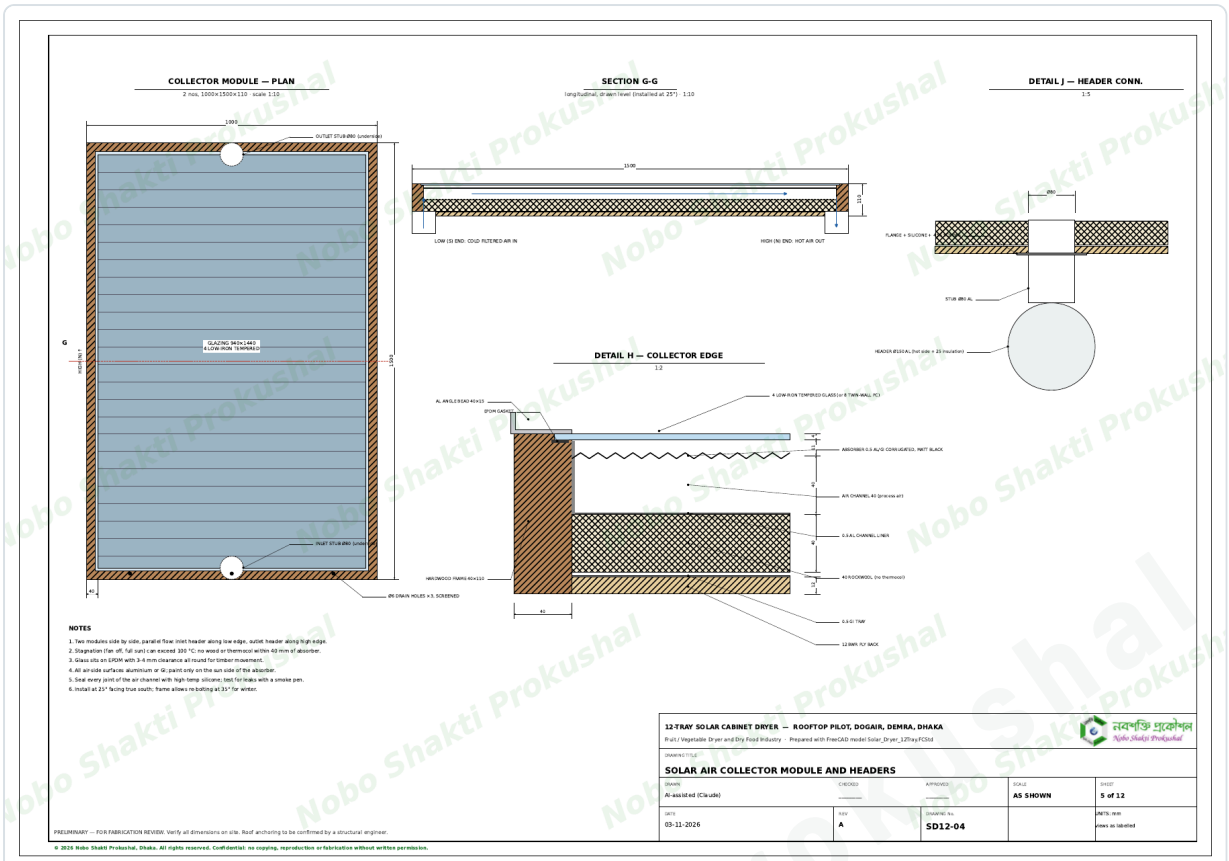


SD12-02 · General arrangement — plan and elevations with overall dimensions

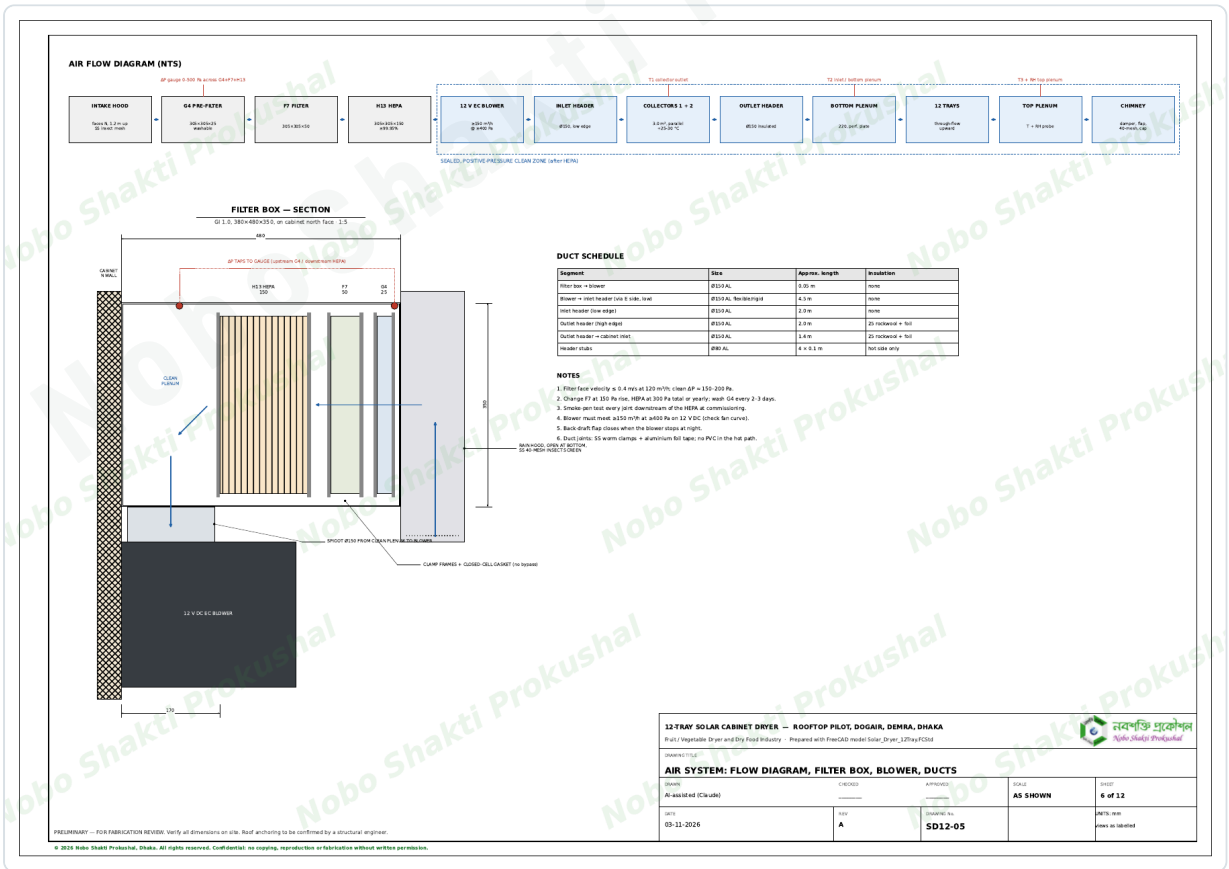


SD12-03 · Drying cabinet — sections, tray stack, wall build-up

M1 · ENGINEERING DRAWINGS



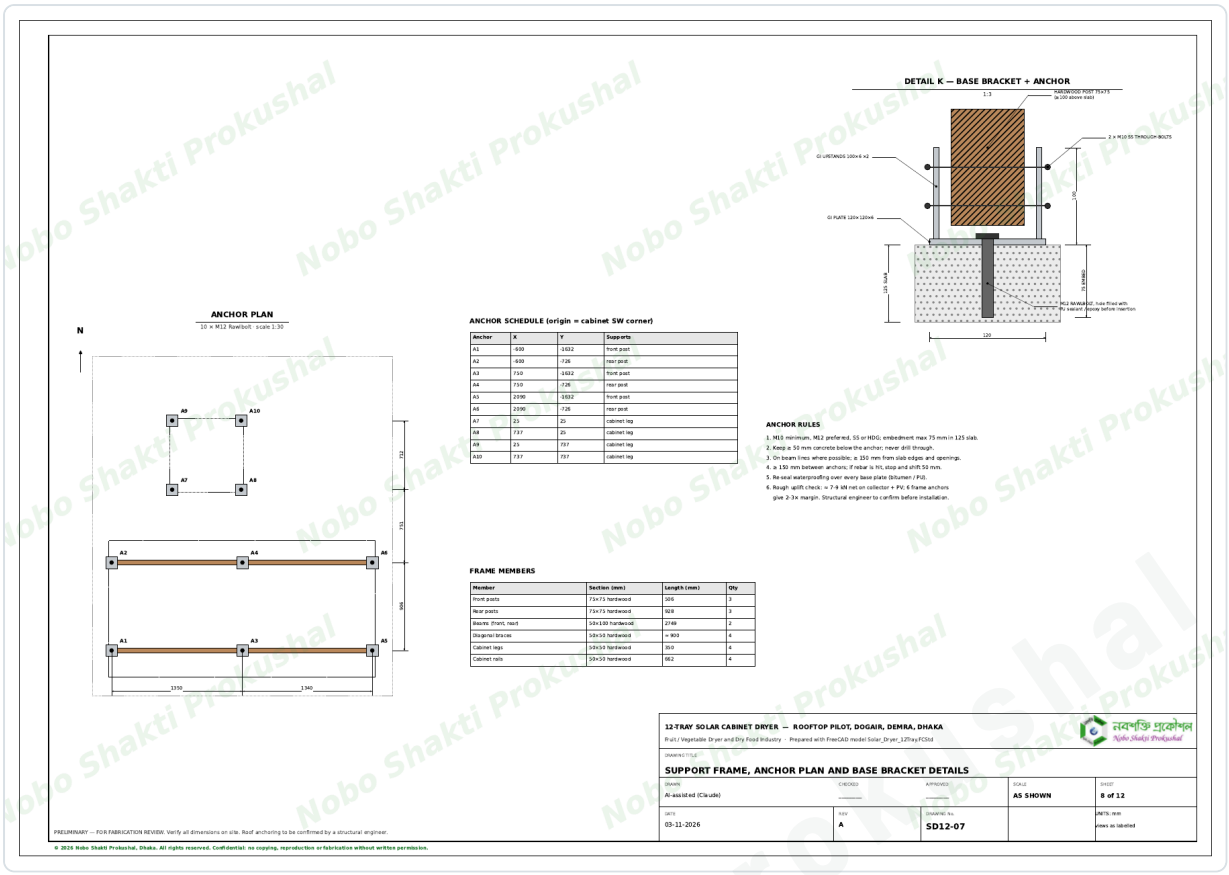
SD12-05 · Solar air collector — back-pass absorber, glazing and headers



SD12-06 · Air path & filtration — G4 / F7 / H13 filter box, blower, ducts

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M1 · ENGINEERING DRAWINGS



SD12-08 · Support frame & anchoring — hardwood stand, GI brackets, Rawlbolt layout

Full M1 documentation package

The complete 12-sheet A3 drawing set (SD12-00 to SD12-11), FreeCAD 3D model, STEP file, AutoCAD DXF drawings and an interactive 3D viewer are available to customers and fabrication partners on request.

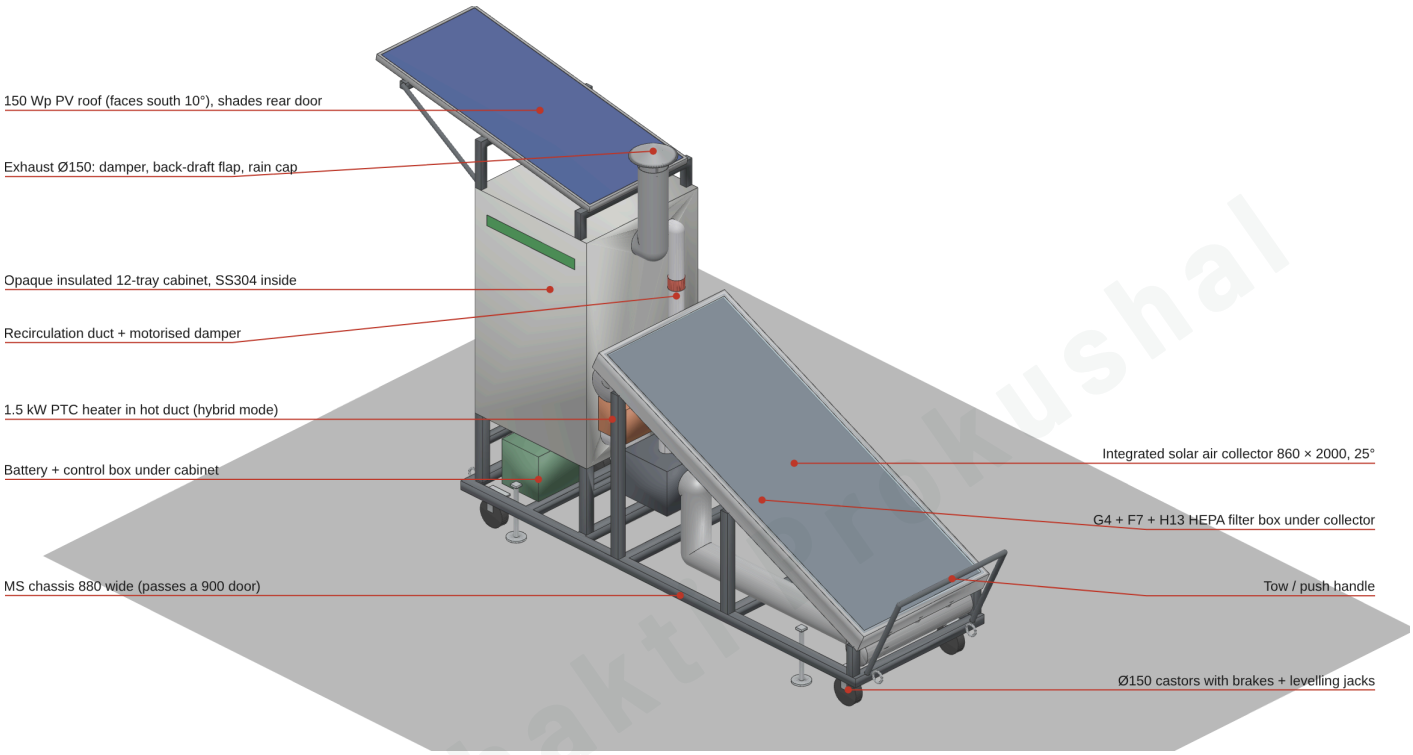
What's in the drawing set

- ✓ Cover, index and general notes
- ✓ General arrangement and site layout
- ✓ Cabinet sections and details
- ✓ Tray and door details
- ✓ Collector, PV frame and filter box
- ✓ Air-flow and electrical schematics
- ✓ Support frame and anchoring
- ✓ Bill of materials
- ✓ Commissioning checklist

MODEL M2 MOBILE SOLAR + HYBRID

NSP-M2 Mobile Solar-Hybrid Dryer

One complete, wheeled unit: collector, HEPA filter box, 12-tray cabinet, PV roof and battery on one steel chassis. Roll it to the orchard, park facing south and start drying — and switch on hybrid heat to keep going through the night and the monsoon.



≈ 16 kg
Fresh slices per 24 h, hybrid*

1.5 kW
PTC heater with safety interlock

880 mm
Width — passes a 900 mm door

12
SS304 trays · 4.9 m²

Best for

- Farm-gate drying, close to the harvest
- Year-round production that can't stop for cloudy days
- Training centres, NGOs and shared village services

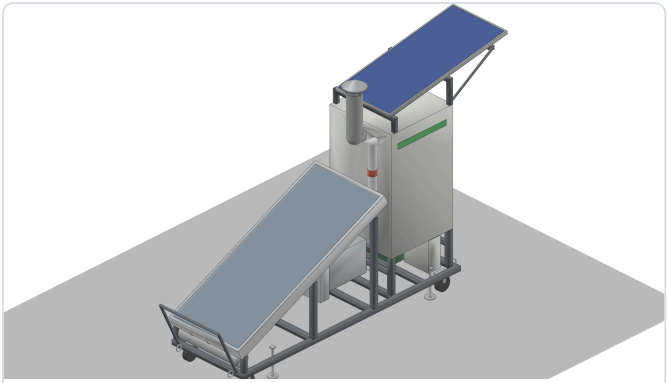
Highlights

- Opaque, PU-insulated cabinet with SS304 liner
- Recirculation duct with motorised damper saves heater energy
- Braked castors, levelling jacks, tie-down eyes, tow handle

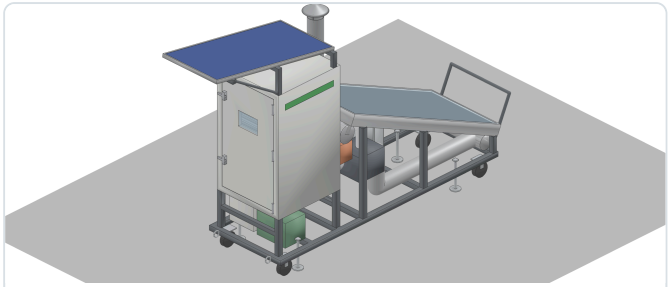
* Design estimates; solar-only output ≈ 5 kg fresh slices per 2 clear days. Hybrid mode needs a 230 V AC supply for the heater. Figures to be confirmed in trials.

M2 · 3D VIEWS

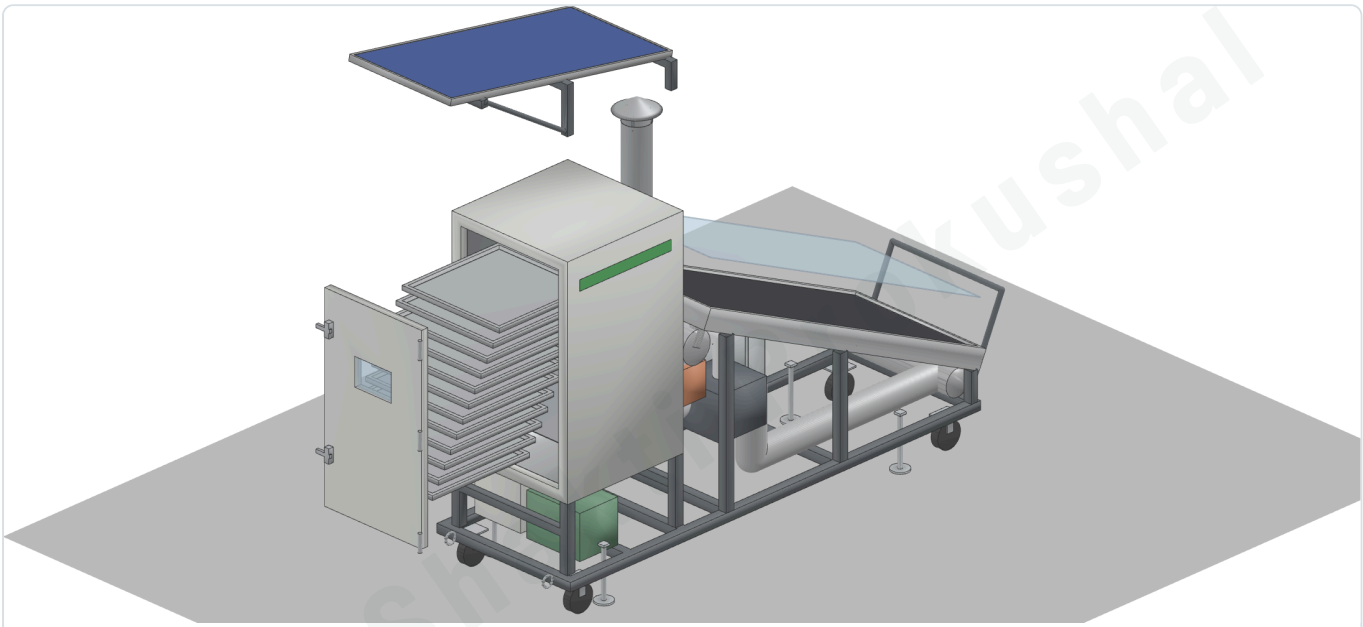
Built as one mobile unit



South-east view — collector end and tow handle



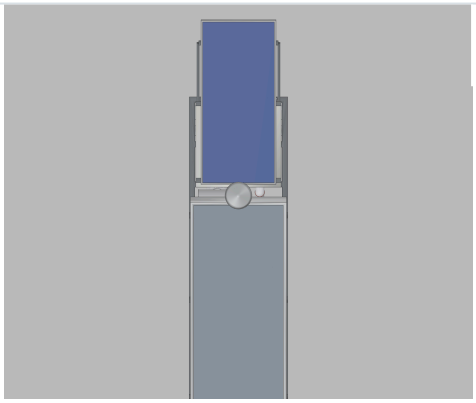
North-west view — rear door under the PV roof



Exploded view — chassis, filter box, collector, cabinet, trays and PV roof



North-east view



Plan view — 3,620 × 880 mm footprint

M2 · TECHNICAL SPECIFICATIONS

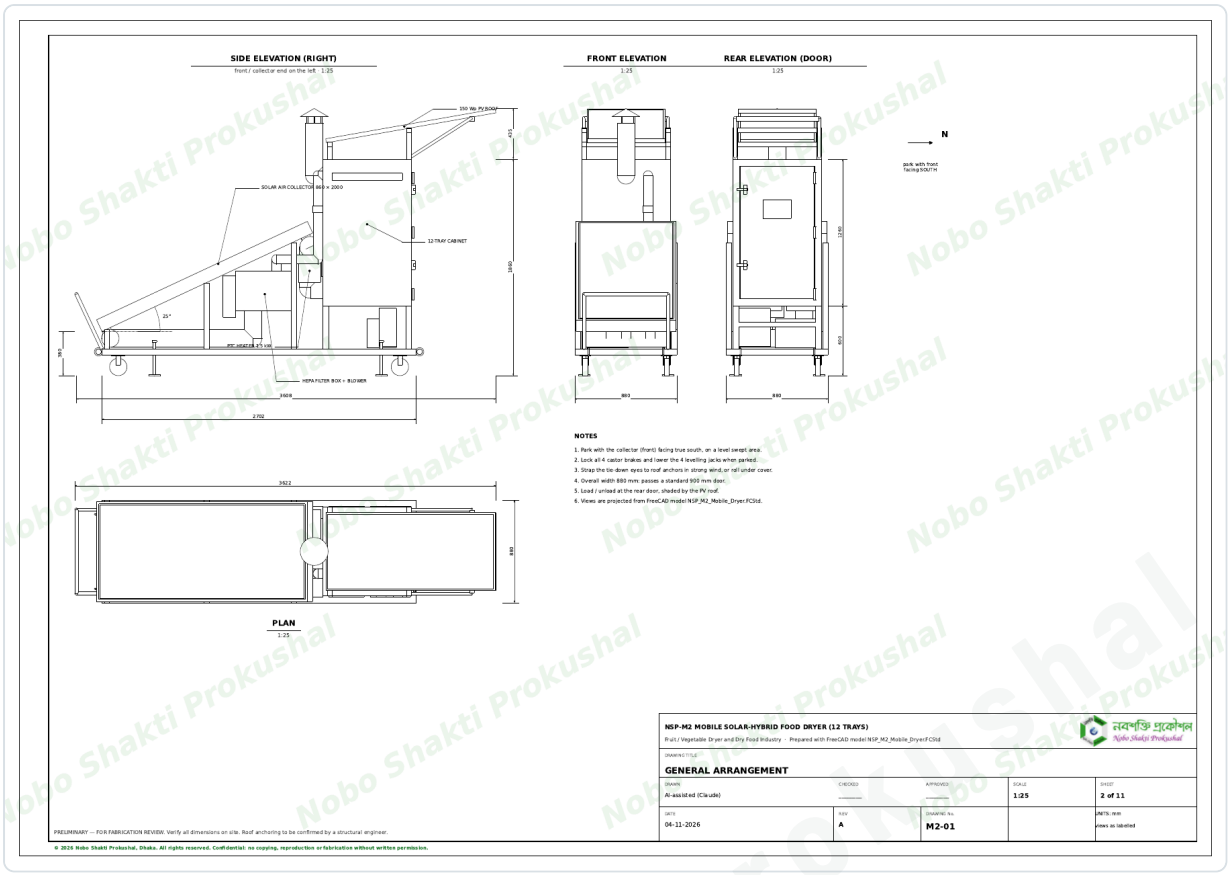
NSP-M2 technical data

ITEM	SPECIFICATION
GENERAL	
Type	Mobile, indirect, fan-forced solar cabinet dryer with electric hybrid heat and recirculation
Overall size (L × W × H)	≈ 3,620 × 880 × 2,300 mm; 880 mm width passes a 900 mm door
Siting	Park on level ground with the collector end facing true south
CHASSIS & MOBILITY	
Chassis	MS SHS 50 × 50 × 2.5 mm, primed and painted
Wheels	4 × Ø150 mm PU castors with brakes
Stability	4 × M24 screw levelling jacks; 4 tie-down eyes; front tow / push handle
DRYING CABINET	
Cabinet	762 × 762 × 1,260 mm, opaque; 1.0 mm GI skin + 40 mm PU + 1.0 mm SS304 liner
Trays	12 × SS304 mesh, 640 × 640 mm, 65 mm pitch — 4.9 m^2
Door	Rear door with small UV-filtered inspection window, silicone gasket
Plenums	Bottom 220 mm with perforated SS distributor; top 180 mm
SOLAR THERMAL	
Collector	Integrated back-pass air collector, 860 × 2,000 × 110 mm (≈ 1.6 m^2 aperture), Al/GI box
Tilt	25°, facing south
AIR HANDLING & FILTRATION	
Filter box	380 × 480 × 340 mm under the collector: G4 → F7 → H13 HEPA with clean plenum
Fan	12 V DC EC blower, ≥ 150 m^3/h, ≥ 400 Pa
Exhaust	Ø150 with damper, back-draft flap and rain cap
Recirculation	Ø80 duct with motorised damper (hybrid / night modes)
HYBRID HEATER	
Heater	1.5 kW, 230 V AC PTC element in the hot down-duct
Safety	Thermostat control, 75 °C over-temperature cut-out, airflow interlock (heater only runs with fan on)
Fuel option	LPG only via a sealed heat exchanger with separate flue — no combustion gas in the food zone
ELECTRICAL & PERFORMANCE	
PV / storage	150 Wp PV roof (south, 10°, shades rear door); MPPT controller and 12 V LiFePO4 battery in a box under the cabinet
Capacity — solar only	≈ 5 kg fresh slices per 2 clear days
Capacity — hybrid	≈ 16 kg fresh slices per 24 h

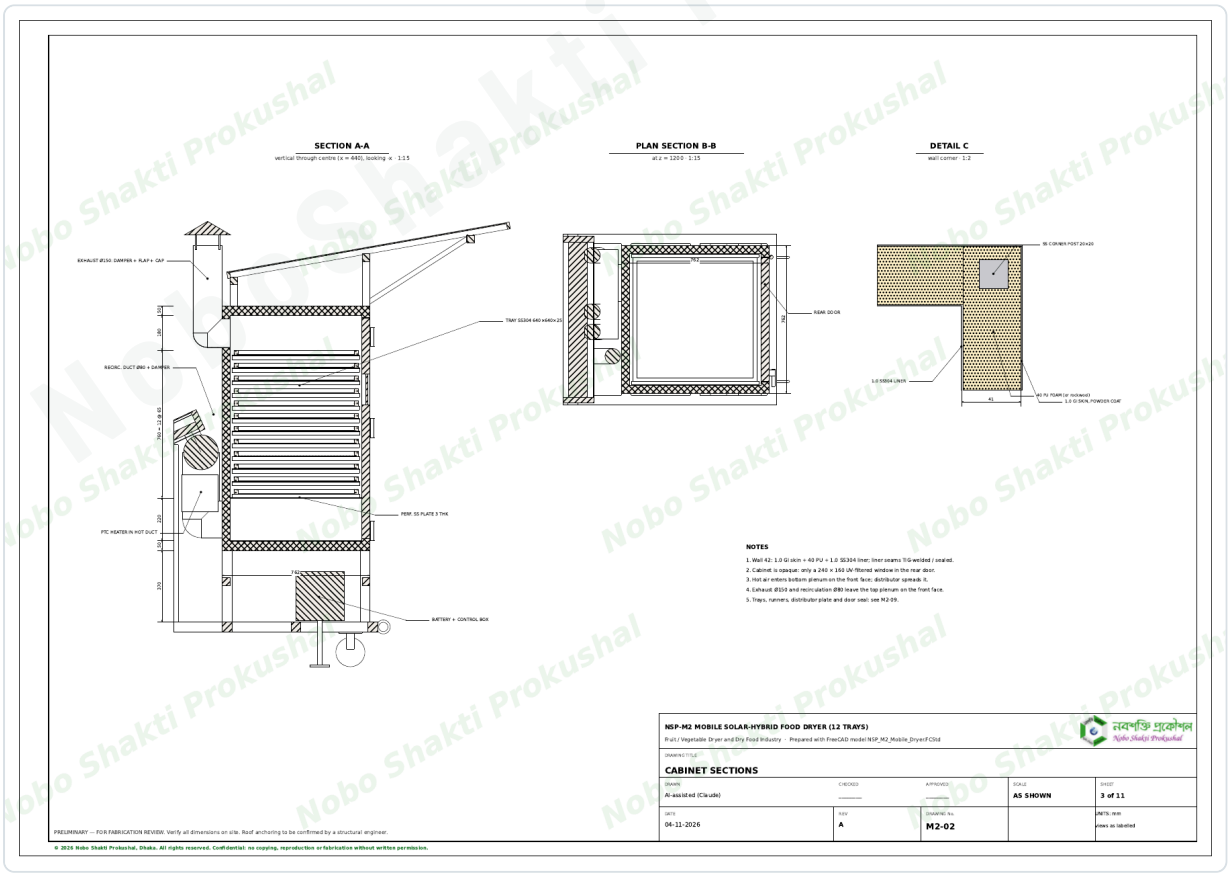
Specifications are for the rev A design and may change as the product is improved. Hybrid electrical installation to be done by a qualified electrician with an earthed, RCD-protected supply.

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M2 · ENGINEERING DRAWINGS

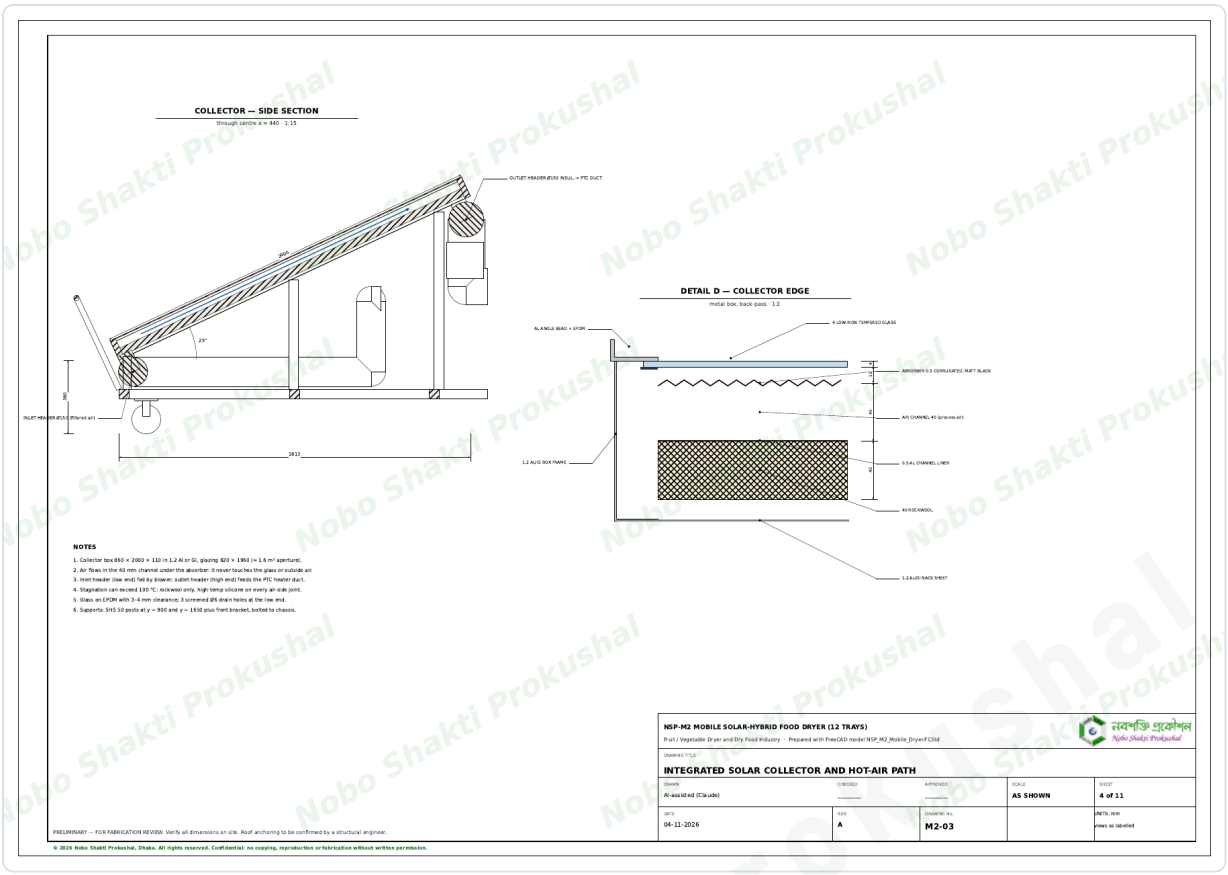


M2-02 · General arrangement — plan and elevations with overall dimensions

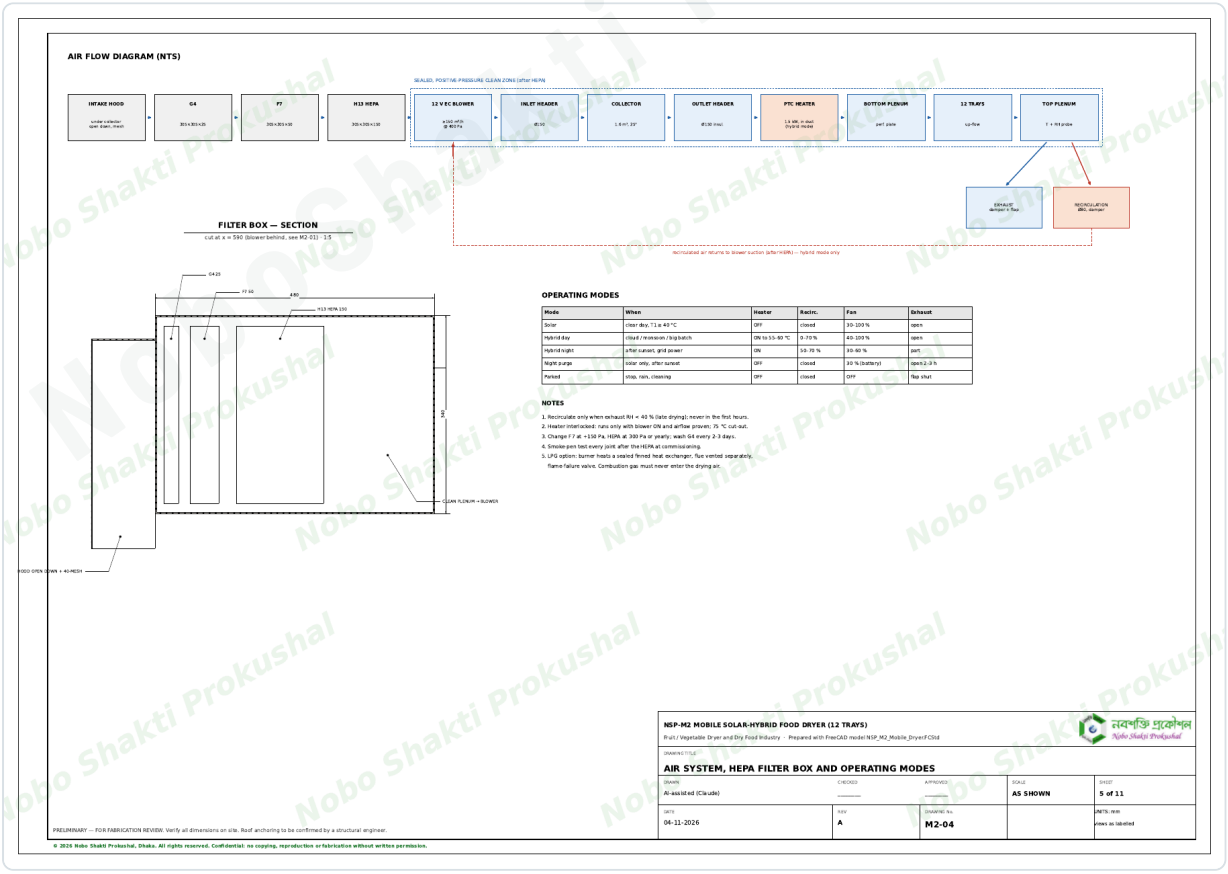


M2-03 · Drying cabinet — sections, plenums, tray stack, wall build-up

M2 · ENGINEERING DRAWINGS

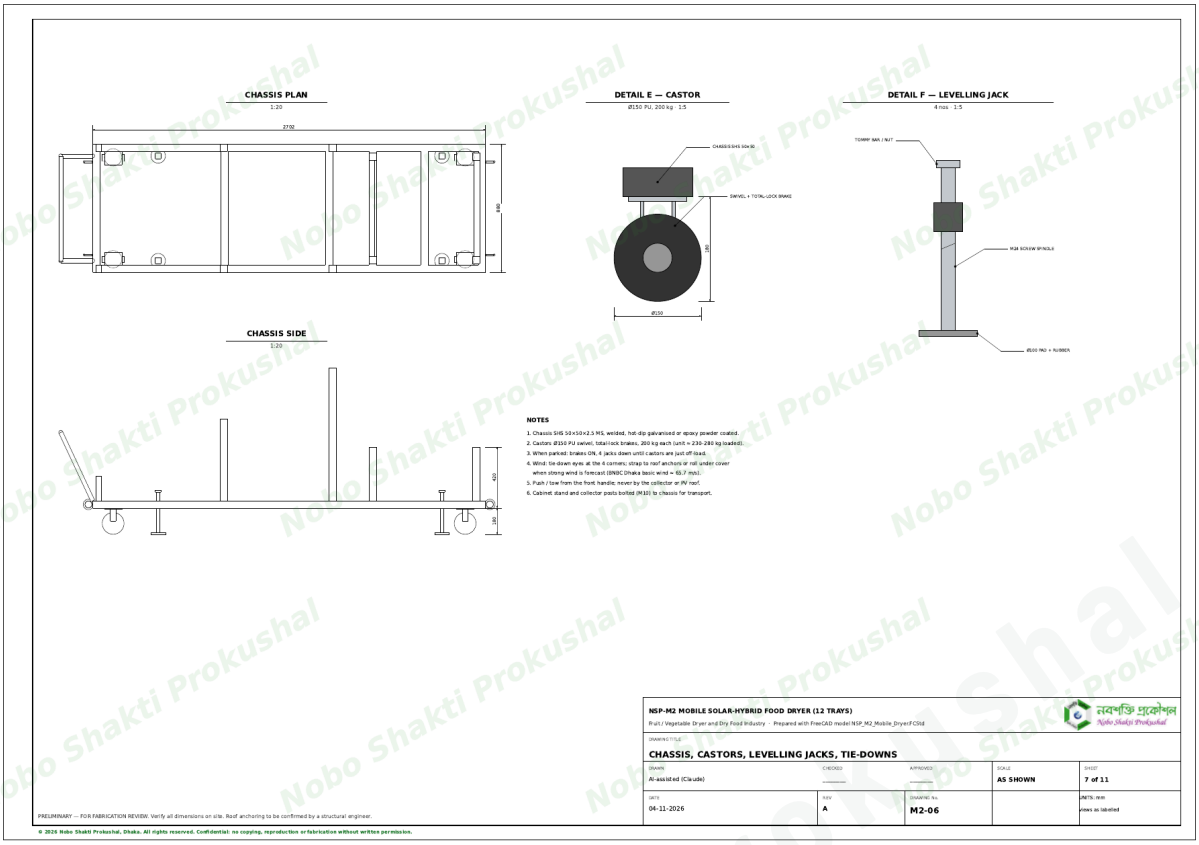


M2-04 · Integrated solar collector



M2-05 · Air path, filtration & hybrid heater

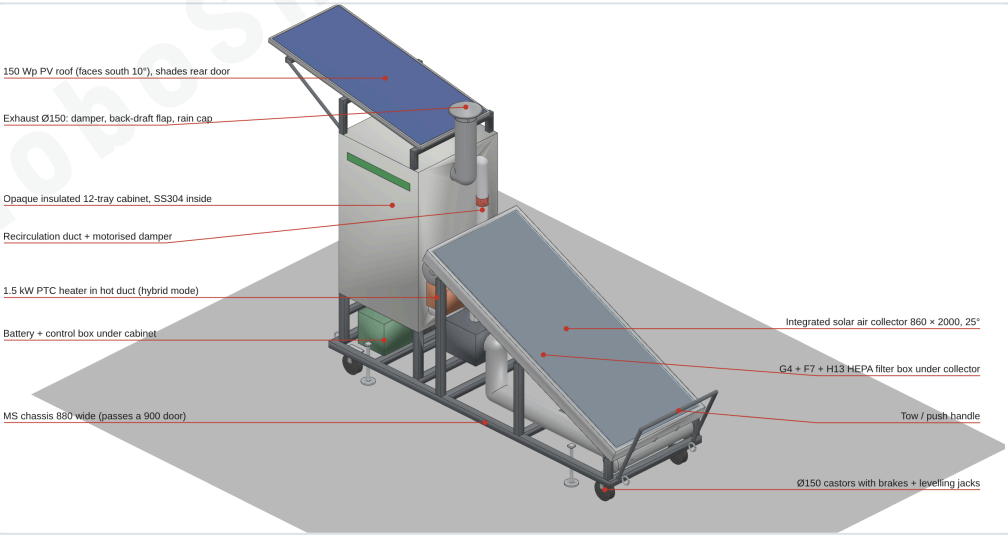
M2 · ENGINEERING DRAWINGS



M2-07 · Mobile chassis — SHS frame, castors, jacks, tie-down points

Full M2 documentation package

The 11-sheet A3 drawing set (M2-00 to M2-10), A1 assembly drawing M2-11 with a 42-item parts list, FreeCAD model, STEP, AutoCAD DXF files and an interactive 3D viewer (explode, section, part info) are available on request.



MODEL SELECTOR

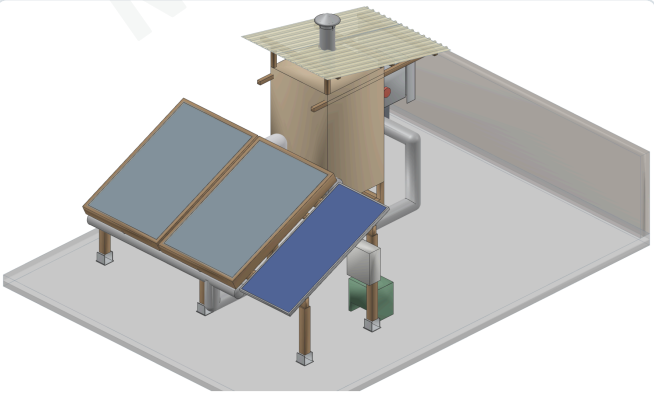
M1 or M2?

FEATURE	NSP-SD12 (M1) ROOFTOP	NSP-M2 MOBILE HYBRID
Installation	Fixed, anchored to roof / slab	Mobile chassis, castors and jacks
Trays / tray area	12 × SS304 · 4.9 m^2	12 × SS304 · 4.9 m^2
Solar collector	3.0 m^2 (2 panels)	≈ 1.6 m^2 (integrated)
Filtration	G4 + F7 + H13 HEPA	G4 + F7 + H13 HEPA
Fan power	12 V DC EC, solar	12 V DC EC, solar
Back-up heat	— (solar only)	1.5 kW PTC + recirculation
Solar-only output*	≈ 8 kg / 2 clear days	≈ 5 kg / 2 clear days
Hybrid output*	—	≈ 16 kg / 24 h
Cabinet shell	Timber + plywood + rockwool + SS liner	GI + PU + SS liner
Space needed	≈ 4.0 × 4.5 m zone	3.62 × 0.88 m unit
Grid power needed	No	Only for hybrid heater
Best for	Rooftop producers, highest solar yield	Farm-gate, all-weather, shared use

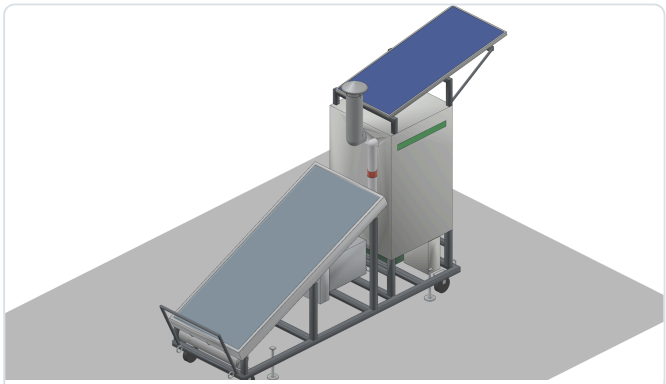
Typical crops and design loads (per 12-tray batch)

CROP	PREPARATION	FRESH LOAD	MOISTURE (WET BASIS)	SOLAR CYCLE (M1)
Mango	5–6 mm slices	≈ 8 kg	≈ 82 → 15 %	≈ 2 days
Pineapple	6 mm rings	≈ 8 kg	≈ 85 → 15 %	≈ 2 days
Banana	5 mm slices	≈ 8 kg	≈ 75 → 15 %	≈ 2 days
Tomato	Halves / slices	≈ 8 kg	≈ 93 → 12 %	≈ 2 days
Green chilli	Whole / split	≈ 6 kg	≈ 85 → 10 %	≈ 2 days
Moringa leaves	Stripped leaves	≈ 5 kg	≈ 75 → 8 %	≈ 1 day

* Design estimates for clear dry-season weather in central Bangladesh. Drying times depend on variety, ripeness, slice thickness, loading and weather; NSP provides a drying guide with every dryer and refines it from trial data.



M1 · Rooftop, highest solar yield



M2 · Mobile, all-weather hybrid

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FOOD QUALITY

Pure product starts on the farm

A clean dryer is only half of the story. NSP is building a supply network that gives dried-food producers better raw material and a product they can sell with confidence.

Hormone-free produce

We work to avoid fruit and vegetables grown or ripened with growth hormones or artificial ripening agents.

Low-chemical farming

Preference for growers who reduce or avoid chemical pesticides, with a clear goal of moving to organic practice.

Organic fertiliser partners

Organic-fertiliser producers are key stakeholders, helping farmers near our dryers get the inputs they need.

Food-grade from day one

SS304 food contact, filtered air, washable surfaces and batch records support international food-safety practice.

Traceable batches

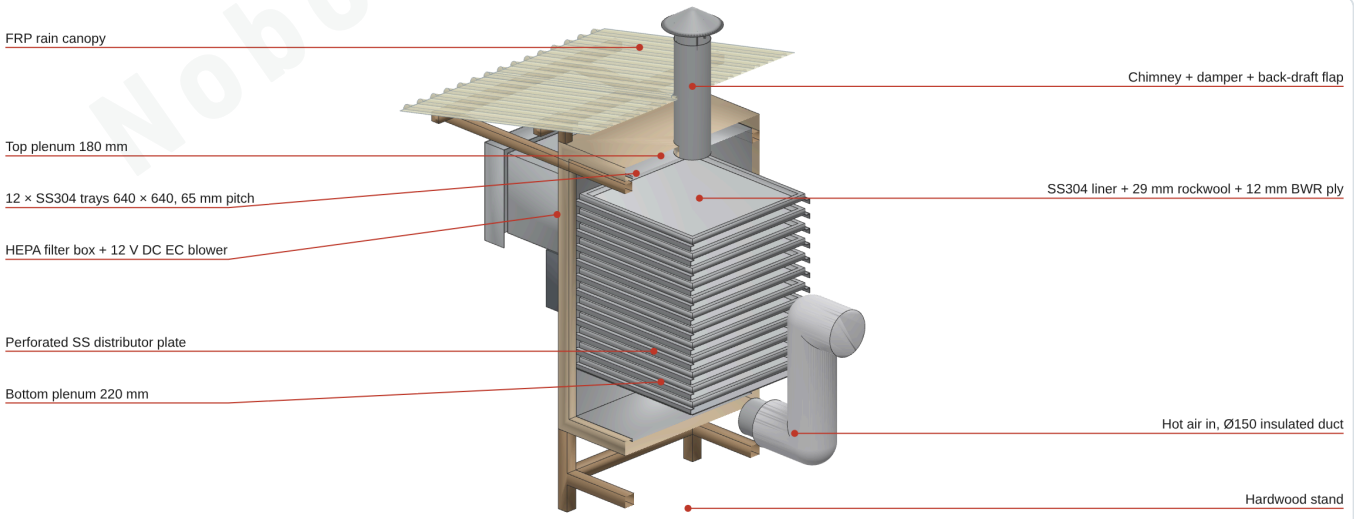
Optional data logging records temperature and humidity for every batch, linked to the source farm.

Fair, local supply chain

Sourcing from areas where each crop is abundant keeps raw-material cost low and income in the community.

The NSP dryer roadmap

STAGE	PRODUCT	STATUS
1	NSP-SD12 (M1) rooftop solar cabinet dryer	Design complete · pilot build
2	NSP-M2 mobile solar-hybrid dryer	Design complete · available to order
3	Heat-pump dryer (closed loop, same 12-tray cabinet)	Preliminary sizing
4	Vacuum freeze dryer for premium products	Planned
5	Food-grade processing plant layout	Planned



Closed, filtered air path — food never meets unfiltered outside air

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COMPLETE SUPPORT

From drawing board to first batch

Services

- **Site survey** — orientation, shading, roof structure and power check
- **Supply & delivery** — fabricated and tested at the NSP workshop
- **Installation** — anchoring, PV wiring, duct sealing
- **Commissioning** — smoke test, airflow and temperature checks
- **Operator training** — loading, hygiene, filter changes, drying guide (Bangla / English)
- **After-sales service** — spares and maintenance from Dhaka
- **Build-it-yourself package** — licensed drawings and supervision for partners

Options & accessories

- Spare filter kit (G4 + F7 + H13 HEPA)
- Extra set of 12 SS304 mesh trays
- ESP32 temperature / humidity data logger
- Larger battery for longer night-hold
- LPG heat-exchanger heater with separate flue (M2)
- Ground stand instead of rooftop frame (M1)
- Custom branding and colour

Pricing

Prices depend on model, options, quantity and site. Contact NSP for a quotation tailored to your project.

PRICE ON REQUEST

How to order

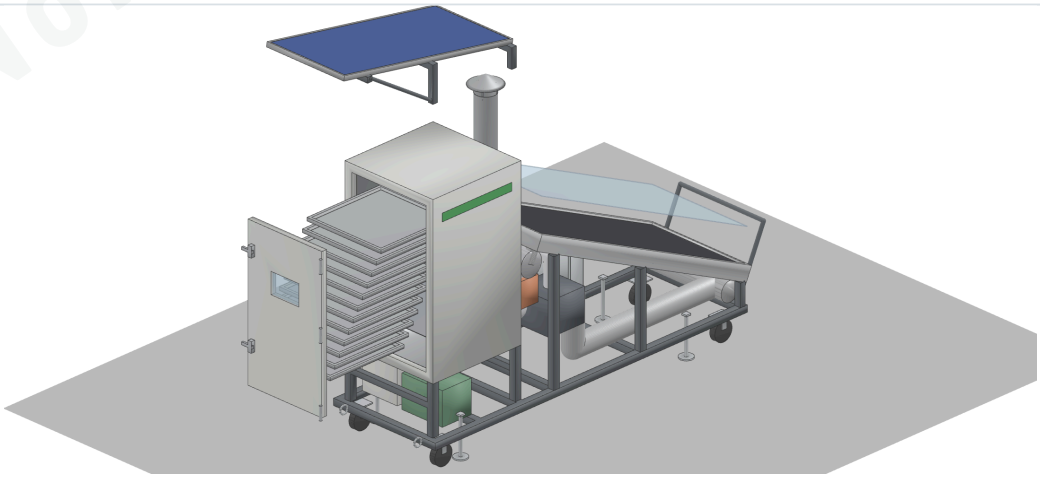
1 ENQUIRE — Call, WhatsApp or email with your crop and site.

2 SURVEY — We check the site and recommend a model.

3 QUOTE — Fixed quotation with delivery schedule.

4 BUILD — Fabrication and factory testing in Dhaka.

5 HANDOVER — Installation, commissioning and training.



Factory-built and tested — every NSP-M2 is assembled and checked at our Khilgaon workshop

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নবশক্তি প্রকৌশল
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Let's dry it clean.

Talk to our engineers about your crop, your site and your production goals.

CORPORATE OFFICE

73/H (GF) Green Road
Dhaka-1205, Bangladesh

WORKSHOP

House #10, Road #5/B
Middle Nandi Para, Khilgaon
Dhaka-1214, Bangladesh

PHONE & WHATSAPP

+880 1714 073604 (WhatsApp)
+880 1711 055476

EMAIL & WEB

info@noboshaktiprokushal.com
www.noboshaktiprokushal.com

Visit our website

Chat on WhatsApp

REGISTRATIONS

Trade Licence 000258
TIN 653549294117
BIN 19021092318
IRC 104713

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